

Article

Multi-View Intrusion Detection Framework Using Deep Learning and Knowledge Graphs

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Abstract: Traditional intrusion detection systems (IDSs) rely on static rules and one-dimensional features, and they have difficulty dealing with zero-day attacks and highly concealed threats; furthermore, mainstream deep learning models cannot capture the correlation between multiple views of attacks due to their single perspective. This paper proposes a knowledge graph-enhanced multi-view deep learning framework, considering the strategy of integrating network traffic, host behavior, and semantic relationships; and evaluates the impact of the secondary fusion strategy on feature fusion to identify the optimal multi-view model configuration. The primary objective is to verify the superiority of multi-view feature fusion technology and determine whether incorporating knowledge graphs (KGs) can further enhance model performance. First, we introduce the knowledge graph (KG) as one of the feature views and neural networks as additional views, forming a multi-view feature fusion strategy that emphasizes the integration of spatial and relational features. The KG represents relational features combined with spatial features extracted by neural networks, enabling a more comprehensive representation of attack patterns through the synergy of both feature types. Secondly, based on this foundation, we propose a two-level fusion strategy. During the representation learning of spatial features, primary fusion is performed of each view, followed by secondary fusion with relational features from KG, thereby deepening and broadening feature integration. These strategies for understanding and deploying the multi-view concept improve the model's expressive power and detection performance and also demonstrate strong generalization and robustness across three datasets, including TON_IoT and UNSW-NB15, marking a contribution of this study. After experimental evaluation, the F1 scores of multi-view models outperformed single-view models across all three datasets. Specifically, the F1 score of the multi-view approach (Model 6) improved by 10.57% on the TON_IoT Network+Win10 dataset compared with the best single-view model. In contrast, improvements of 5.53% and 3.21% were observed on the TON_IoT network and UNSW-NB15 datasets. In terms of feature fusion strategies, the secondary fusion strategy (Model 6) outperformed primary fusion (Model 5). Furthermore, incorporating KG-based relational features as a separate view improved model performance, a finding validated by ablation studies. Experimental results show that the deep fusion strategy of multi-dimensional data overcomes the limitations of traditional single-view models, enables collaborative multi-dimensional analysis of network attack behaviors, and significantly enhances detection capabilities in complex attack scenarios. This approach establishes a scalable multimodal analysis framework for intelligent cybersecurity, advancing intrusion detection beyond traditional rule-based methods toward semantic understanding.



Academic Editor: Jian Tang

Received: 12 March 2025

Revised: 24 April 2025

Accepted: 27 April 2025

Published: 1 May 2025

Citation: Li, M.; Qiao, Y.; Lee, B. Multi-View Intrusion Detection Framework Using Deep Learning and Knowledge Graphs. *Information* **2025**, *16*, 377. <https://doi.org/10.3390/info16050377>

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Keywords: deep multi-view learning (DMVL); anomaly detection; knowledge graph (KG)

1. Introduction

In recent years, intrusion detection systems (IDSs) have emerged as a critical security defense mechanism, effectively identifying and mitigating malicious intrusions [1,2]. These systems can be broadly categorized into network intrusion detection systems (NIDSs) and host-based intrusion detection systems (HIDSs). Specifically, the primary task of an NIDS is to continuously monitor network activity and determine, in real time, whether an alert should be triggered for the system administrator, while an HIDS focuses on analyzing logs, system behaviors, and file integrity to detect suspicious activities. Despite their widespread adoption, traditional IDS exhibit critical drawbacks such as high false positive/false negative rates, computational overhead, and reliance on domain knowledge. A robust IDS, whether it be network- or host-based, must be able to automatically identify attack behaviors or potential threats concealed within monitored data, ensuring stable and efficient operation of systems within both network and host environments. To enhance protection against cyber threats, researchers have focused on developing more sophisticated feature extraction techniques and efficient detection models.

In the early stages of the Internet, network attacks were relatively limited in scope and could often be countered effectively through port-based detection mechanisms and firewall technology [3]. However, as network behaviors have become increasingly intricate, involving various communication protocols and varied user activities, traditional port-based and rule-based detection methods have shown significant limitations. Port-based detection fails against port-hopping attacks due to its reliance on static port mappings, while rule-based systems suffer from labor-intensive signature updates, resulting in unsustainable maintenance overheads and inadequate protection against zero-day attacks, including polymorphic malware. These methods also struggle to extract precise features and suffer from high false positive rates when dealing with massive data flows and sophisticated attack patterns characteristic of modern networks [4]. Furthermore, traditional anomaly detection approaches are increasingly inadequate in addressing the complexity of new and evolving attack scenarios due to their limited capabilities in feature extraction and model development.

However, anomaly detection methods typically suffer from a high false positive rate, which limits their reliability [4]. To solve this problem, this study uses deep learning techniques to enhance the intelligence of anomaly detection systems, ultimately improving the precision of network intrusion detection.

Traditional deep learning models either use single-dimensional traffic features or simple neural networks. These approaches are generally inadequate for capturing the intricate relationships within multi-source heterogeneous data, resulting in feature omission and insufficient detection accuracy. To overcome these challenges, some studies have increasingly focused on multi-view learning, aiming to capture effective features from heterogeneous data in a more comprehensive manner. Multi-view models enable a detailed representation of network behaviors by integrating information from various sources. For example, network traffic data can be observed from various perspectives, with features extracted and subsequently fused. Combining network traffic features with host-level activity data allows for a more nuanced depiction of an attacker's behavior, including attempts to traverse network boundaries. This integrative approach ultimately enhances the effectiveness and comprehensiveness of intrusion detection systems.

In multi-view models, deep learning techniques are often employed to unilaterally extract features (such as spatial or relational characteristics). For instance, convolutional neural networks (CNNs) are frequently used to capture spatial features, while recurrent neural networks (RNNs) are effective at modeling local temporal relationships [5,6]. However, these approaches tend to fall short of fully exploring the global complementarity inherent amongst different views. In this context, the current study proposes incorporating KG as an additional feature view to facilitate the integration of spatial and relational features. KG provides a structured framework by linking entities and events, thereby enabling a systematic representation of network traffic data [7]. Within network intrusion detection, KG is particularly useful for describing the complex associations between network entities (e.g., IP addresses, ports, hosts), enriching the model's comprehension of semantic context, and ultimately enhancing the ability of multi-view models to identify latent threats [8].

The primary objective of this study is to verify the superiority of multi-view features fusion technology in intrusion detection and to determine whether incorporating KGs can further improve model performance. Moreover, the study evaluates the impact of first-level and second-level fusion strategies on feature integration to identify the optimal multi-view model configuration.

The main contributions of this paper are summarized as follows:

- (1) **Multi-view features fusion to enhance anomaly detection:** This study presents a multi-view data fusion framework that integrates host data and network traffic data to improve anomaly detection performance. The robustness and generalizability of the proposed model were evaluated using the TON_IoT and UNSW-NB15 datasets.
- (2) **Diversified model evaluation for intrusion detection:** This study conducted a comprehensive evaluation of both single-view models (CNN, LSTM, CNN+attention LSTM) and multi-view fusion models ((KG+CNN)+attention LSTM, (KG+Multi-CNNs)+attention LSTM). The results highlight the advantages of multi-view fusion in enhancing detection accuracy and reliability over traditional single-view approaches.
- (3) **Optimization of multi-view fusion strategies and the role of KG:** This study assessed the impact of one-level and two-level fusion strategies on model performance under consistent experimental conditions. Through ablation studies, the contribution of KG to feature fusion and anomaly detection was validated, emphasizing the pivotal role of KG in boosting the effectiveness of multi-view models.

The remainder of this paper is organized as follows: Section 2 provides background information and reviews related work in the field of network intrusion detection. Section 3 presents the proposed methodology in detail, including the design of the model and the construction of the KG. Section 4 outlines the preparation steps for the experiments, including a brief description of the TON_IoT network, TON_IoT network+Win10, and UNSW-NB15 datasets used in the study, as well as data preprocessing procedures for the experimental evaluation and integration of the multi-view framework. Section 5 details the experimental setup and results, including the deployment of single-view and multi-view models as well as ablation experiments. Finally, Section 6 summarizes the experimental conclusions and discusses the practical application value of the proposed model.

2. Related Work

In line with the focus of this study, we provide a concise review of related work in the areas of intrusion detection systems based on single-view and multi-view models and cybersecurity KGs.

In the field of intrusion detection, the current mainstream methods are based on machine learning (ML) or deep learning (DL). For example, k-nearest neighbor (KNN) [9] and support vector machine (SVM) [10] have shown considerable success in addressing

classification and clustering tasks using well-known datasets like KDD99, NSL-KDD, and DARPA, following a single-view strategy. However, these datasets are now largely considered obsolete, as they contain overly simplistic attack patterns that do not accurately reflect the complexity of today's network environments [11].

To address the limitations inherent in single-view intrusion detection systems, multi-view models have been developed to fuse and embed data from multiple perspectives into a shared latent space, thereby facilitating the extraction of richer and more informative representations. One representative traditional approach is canonical correlation analysis (CCA) [12] which, while offering some capability in fusing data, is limited in capturing high-order interactions across heterogeneous data sources. In response to these limitations, a variety of deep learning-based architectures have emerged, including multimodal deep Boltzmann machines [13], multimodal deep autoencoders [14], and multimodal recurrent neural networks [15]. The crux of multi-view representation learning lies in its ability to model intricate dependencies and interactions between multiple data views. This representation learning method has shown its superiority in other studies. For example, Peng et al. [16] proposed a deep multi-view framework that leverages multi-view learning in combination with graph convolutional networks (GCNs) to tackle anomaly detection in attributed networks. Comparative experiments indicated that this framework outperforms a range of existing baseline models, including both traditional single-view and multi-view methods, demonstrating the effectiveness of the multi-view approach in complex network settings.

As a security-specific KG, the cybersecurity KG comprises nodes and edges that form an extensive security semantic network, enabling the modeling of various real-world attack and defense scenarios. Beyond information extraction (IE) [17,18], cybersecurity KG presents several advantages, such as entity disambiguation [19], enabling the effective extraction and integration of knowledge from heterogeneous data from multi-sources. Furthermore, cybersecurity KG facilitates the structured representation of cybersecurity knowledge, capturing relationships between entities, and visualizing this information graphically, which provides an efficient and highly intuitive means of comprehension [20].

In addition to the intrusion detection methods discussed in [21], KGs have demonstrated constructive potential in supporting intrusion detection efforts. Yang et al. [22] proposed a DDoS detection method based on KG technology, extracting key feature pairs related to network attacks through semantic feature analysis and co-occurrence calculations, effectively capturing semantic relationships between traffic characteristics. In their work, the KG is used to represent the communication processes between hosts. By integrating multi-view feature fusion, they combined features extracted from the knowledge graph with statistical analysis features, enabling efficient detection and classification of threats. Garrido et al. [23] applied machine learning techniques to KG to detect anomalous activities in industrial automation systems that integrate IT and OT components. Xiao et al. [24] proposed a KG embedding method to predict intra-type and inter-type relationships among software security entities, thereby assisting analysts in enriching software security knowledge by identifying previously unknown relationships. However, it should be noted that the cybersecurity KG described by Xiao et al. [24] is not open-source, restricting access to its underlying details. The KG part of this study is constructed by integrating three main sources of knowledge: host-level information, network-level observations (e.g., inter-host connections), and application-level observations (e.g. data access events) [25]. This approach aims to create a comprehensive cybersecurity KG that facilitates a better understanding of network behaviors.

Upon reviewing current research on single-view models, multi-view models, and KG within the domain of network security for intrusion detection, it becomes evident that most existing methods tend to rely on a single technical paradigm. In the related studies mentioned above, either independent traditional neural network models or isolated KGs were used. While these techniques have achieved noteworthy results in their respective domains, Garrido et al. [23] mentioned that traditional IDS detects events independently and lacks a holistic view of complex attack patterns. At the same time, it is noted that the knowledge graph method may be combined with the traditional intrusion detection system (IDS) in the future, but no specific implementation path has been given. Even when single-technology neural network models are enhanced using multi-view fusion techniques, the fusion is often restricted to the extension of spatial features, failing to fully harness the comprehensive potential of multi-source information. Likewise, while KGs effectively capture the relational characteristics between network entities, they are unable to comprehensively represent and integrate spatial features. Feature representation in intrusion detection is inherently multi-dimensional, encompassing spatial, relational, and temporal aspects; however, current approaches struggle to effectively integrate these diverse features.

In response to these limitations, our study advances the concept of multi-view models at two distinct levels. First, we fuse KGs and neural network as different views to achieve effective integration of multi-view features. This multi-view features fusion strategy emphasizes the combination of spatial and relational features—representing relational characteristics through KG and integrating them with spatial features extracted by neural networks. This synergistic integration provides a more comprehensive representation of attack patterns, thus significantly enhancing the accuracy and robustness of intrusion detection.

Building on this foundation, we introduce a two-level fusion strategy. At the first level, spatial features are fused within each view, followed by a second-level fusion that integrates these features with the relational information captured in the KG. This hierarchical fusion enhances both the depth and breadth of feature integration, resulting in richer, more expressive feature representations. The proposed two-level fusion strategy not only improves the model's detection performance but also demonstrates superior generalization and robustness across multiple datasets (such as TON_IoT and UNSW-NB15). These contributions represent a key highlight of the study, offering a novel approach to tackling the challenges associated with multi-dimensional feature representation in network intrusion detection.

3. Methodology

3.1. Overview

This study has two objectives: (i) to determine whether a multi-view fusion-based model outperforms single-view models in intrusion detection within network security; and (ii) to assess whether incorporating a KG as an additional feature view can enhance the multi-view model's performance. The following sections outline the approaches used in the study to achieve its goals.

Baseline Analysis of Single-View Models: To establish baselines for our multi-view framework, we developed three distinct single-view models, each with a different architecture: CNN, LSTM, and a hybrid model that combines CNN with attention-based LSTM. These single-view models were chosen mainly because the multi-view fusion model is designed based on these single-view models. Such a comparison helps to verify the effectiveness and contribution of the fusion strategy from an experimental perspective. These single-view models serve as baseline comparisons, and we use them to evaluate the effectiveness of our multi-view fusion strategies.

Multi-View Features Fusion Strategy Evaluation: The multi-view fusion approach employed in this study integrates host-based features and network traffic data through a spatial and relational lens. More concretely, our methodology involves the fusion of spatial features extracted through a CNN with relational features derived from a KG. We explored two distinct fusion strategies:

Fusion Strategy 1— This approach integrates features extracted from both the KG and the CNN, thereby combining relational and spatial characteristics into a unified representation.

Fusion Strategy 2—This approach extends the multi-view learning capability by integrating multiple CNN modules with the KG to achieve a layered, hierarchical fusion. On the one hand, the CNN module extracts local features of each perspective in multi-view fusion, reflecting the spatial features of different perspectives. On the other hand, KG provides explicit relational features from a global perspective. The associations, constraints, and logical relationships between entities are integrated into the feature expression. This makes up for the important features that are missed due to the associations between local features. This two-layer fusion mechanism not only effectively combines local spatial features with global relational features, but also makes up for the shortcomings of a single perspective in feature expression; KG can help to make up for the missing information between local features and more comprehensively explores the synergy between spatial features and relational features.

KG Ablation Experiment: To assess the contribution of relational features from KG in our multi-view fusion strategy, we performed ablation experiments. By comparing the performance of the complete model—including the KG component—with a variant excluding this component, we observed the impact of KG-based features on the overall effectiveness of feature fusion and anomaly detection.

To validate the robustness of our models and to assess their generalizability across heterogeneous network and host environments, we conducted experiments utilizing three distinct datasets: TON_IoT network+Win10, TON_IoT network, and UNSW-NB15. These datasets allow for a comprehensive evaluation of the proposed models, ensuring that their efficacy can be demonstrated under varied and complex network security scenarios.

3.2. Model Design

In this section, we introduce the design structure of all the models used in the experiment. To simplify the naming, each model is numbered.

Model 1: CNN. In this study, a single-view CNN is utilized to process three raw datasets directly without involving traditional feature engineering techniques. This single-view CNN serves as a baseline for comparison with the more advanced multi-view CNN model that will be introduced later. The CNN follows a conventional architecture, comprising an input layer, convolutional layers, pooling layers, and fully connected layers [26,27].

Model 2: LSTM. LSTM is a specialized recurrent neural network (RNN) designed to address the vanishing and exploding gradient issues in traditional RNNs during long-term dependency tasks. The LSTM model architecture in this study consists of an input layer, an LSTM layer, fully connected layers, and an output layer. The LSTM layer, through its memory cells (cell states), input gate, forget gate, and output gate, effectively captures complex temporal dependencies present in network traffic analysis. This study employs a single-view LSTM model, which means it only receives input from a single feature space [26,27].

Model 3: CNN+Attn-Based LSTM. This model integrates CNN, LSTM, and attention mechanisms to effectively capture both spatial and temporal features inherent in network traffic data. Initially, a CNN is utilized to process the complete feature set from all datasets, uncovering local dependencies and essential patterns in the data. This single-view model establishes a foundational benchmark for comparison against more sophisticated multi-view frameworks that incorporate KG for enriched feature representation. The spatial features derived from CNN are subsequently passed to an LSTM layer, enabling the model to learn the temporal relationships embedded in the data sequences. To improve the model's ability to highlight key temporal dependencies, an attention mechanism is integrated. Finally, a classification layer leverages these enhanced features to classify network traffic data. By combining spatial and temporal insights, this architecture is well equipped to capture complex patterns and dependencies that distinguish between typical and atypical network behaviors.

Model 4: multi-CNNs+Attn-Based LSTM. The model introduced in this section employs a multi-convolutional network structure during the feature extraction phase to better capture the diversity of different feature spaces. This is another baseline framework. Later, a multi-view framework will be built based on this baseline framework by adding multiple views of KG. Specifically, the model initially divides each dataset into multiple subsets, with each subset passing through an independent convolutional layer to extract spatial features from various perspectives. This multi-perspective convolutional structure enables the model to capture more diverse feature expressions, thereby providing a comprehensive understanding of the complex feature space. Once the spatial features are extracted, they are fused and fed into an LSTM layer equipped with an attention mechanism. This configuration is intended to learn temporal dependencies while emphasizing the most significant features. Ultimately, the fused features are classified by a fully connected layer to determine whether the network traffic is normal or anomalous.

Model 5: (KG+CNN)+Attn-Based LSTM. The proposed multi-view KG+CNN+attention LSTM model in this study effectively facilitates the comprehensive detection of network traffic anomalies through multi-stage feature extraction and fusion. The framework, aside from data preprocessing, is composed of four components: feature extraction through two complementary models (KG+CNN), view pooling, attention-based modeling, and final classification.

Initially, during the data preprocessing stage, data cleaning and feature encoding are conducted on each dataset to ensure data quality, uniformity, and reliability. In the subsequent feature extraction stage, relational features are derived from KG, while spatial features are captured using CNN. The KG is employed to reveal intricate inter-feature relationships, whereas the CNN is utilized to extract significant local patterns present in the network traffic. This strategy integrates spatial and relational features in the fusion layer to achieve a more comprehensive feature representation .

$$\mathbf{F}_{\text{fused}} = \sigma(\mathbf{W}_1 \cdot [\mathbf{F}_{\text{CNN}} \parallel \mathbf{F}_{\text{KG}}]), \quad (1)$$

Here, $\mathbf{F}_{\text{CNN}} \in \mathbb{R}^{d_1}$ denotes the spatial features extracted by the CNN module, and $\mathbf{F}_{\text{KG}} \in \mathbb{R}^{d_2}$ denotes the relational features derived from the KG.

Figure 1 shows the details of the framework based on the TON_IoT network and UNSW-NB15 datasets. Among them, the process of generating KG is further shown in Figure 2.

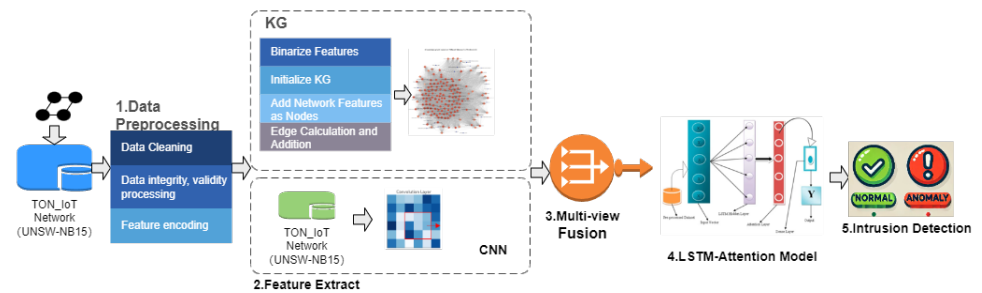


Figure 1. (KG+CNN)+attention LSTM multi-view model based on the TON_IOT network (UNSW-NB15).

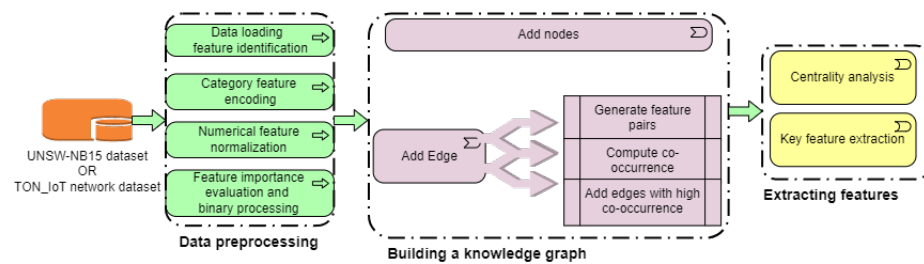


Figure 2. Construction of the KG based on the TON_IOT network (UNSW-NB15).

For the combined TON_IoT network and Win10 dataset, the feature extraction approach involves some notable adjustments, primarily concerning the CNN-based feature extraction process. Figure 3 shows the framework details based on the TON_IoT network+Win10 combined dataset. Also, the KG process framework as a part of the model is further illustrated in Figure 4.

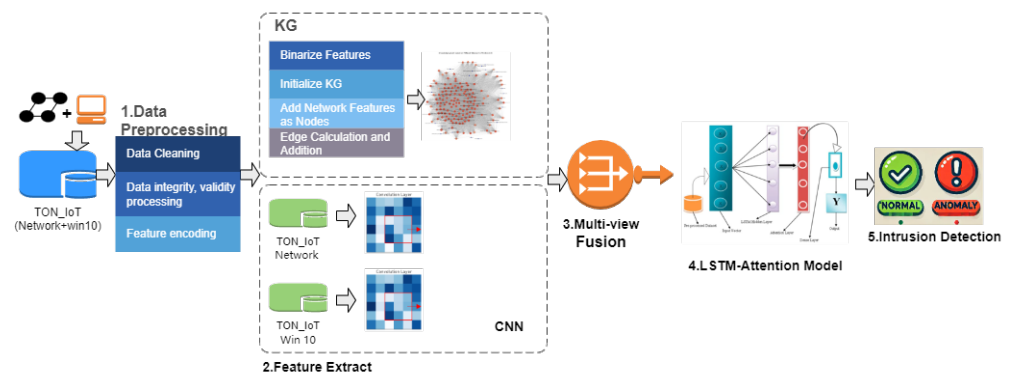


Figure 3. (KG+CNN)+attention LSTM multi-view model based on TON_IOT network+Win10.

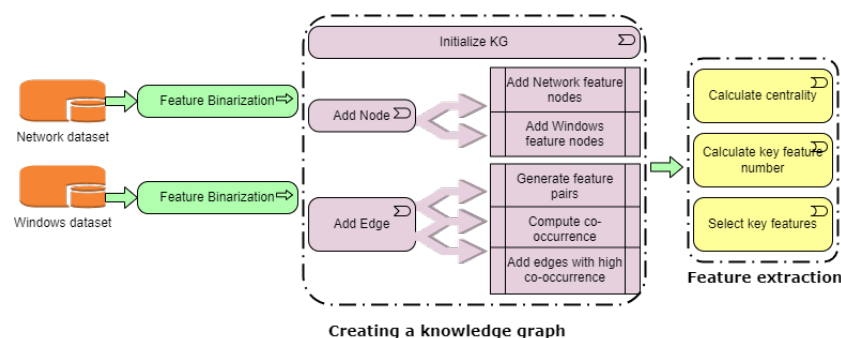


Figure 4. Construction of the KG based on TON_IOT network+Win10.

The model processes the TON_IoT network and Win10 sub-datasets independently and then merges the features in the fusion layer of CNN (Step 2: Feature Extract). The combined CNN features are then further integrated with the relational features derived from KG (Step 3: Multi-View Fusion). This fused feature representation is then passed to the LSTM layer equipped with an attention mechanism (Step 4), which aims to capture temporal patterns while emphasizing key features. Finally, the fully connected classification layer is used to determine whether network traffic is normal or anomalous. This collaborative mechanism allows the model to effectively leverage the multi-view information from different datasets, thereby enhancing the accuracy and robustness of network traffic anomaly detection.

Model 6: (KG+multi-CNN)+Attn-Based LSTM. This model further extends the multi-view learning framework, aiming to better manage data diversity and complexity, thus enhancing network traffic anomaly detection performance. Here, the feature extraction stage is different. This model implements a two-level fusion approach. In the first level, the dataset is divided into several subsets from different perspectives (See Section 4.2 for details on multi-view subsets), and each subset is processed independently by convolutional layers. These extracted features are then integrated through view pooling, which ensures comprehensive utilization of the feature expressions from each perspective. In parallel, relational features are extracted from the KG. The following is the expression of the first level fusion (local spatial features). For N CNN modules (multi-view), let $\mathbf{F}_{\text{CNN}}^{(i)}$ be the features from the i -th CNN. These are first fused via weighted aggregation:

$$\mathbf{F}_{\text{local}} = \sum_{i=1}^N \alpha_i \mathbf{F}_{\text{CNN}}^{(i)}, \quad (2)$$

where α_i is an attention weight for the i -th view, learned via:

$$\alpha_i = \frac{\exp(\mathbf{v}^T \mathbf{F}_{\text{CNN}}^{(i)})}{\sum_{j=1}^N \exp(\mathbf{v}^T \mathbf{F}_{\text{CNN}}^{(j)})}, \quad (3)$$

with $\mathbf{v}$ as a learnable vector.

In the second level, the fusion process integrates these spatial features with the relational features derived from KG, thereby enriching the overall feature representation. The resulting fused features are subsequently input into an LSTM layer with an attention mechanism. During the classification stage, the fully connected layer is used to classify the fused features, determining whether the network traffic is normal or anomalous. The following is the expression of second-level fusion (global relational+local features). The local features $\mathbf{F}_{\text{local}}$ are integrated with KG features $\mathbf{F}_{\text{KG}}$ via:

$$\mathbf{F}_{\text{fused}} = \sigma(\mathbf{W}_2 \cdot [\mathbf{F}_{\text{local}} \parallel \mathbf{F}_{\text{KG}}]), \quad (4)$$

The structure of the model is shown in Figures 5 and 6. The focus is on how to improve the detection performance through the synergy of multi-CNN modules and KG. The KG process framework as part of the model is shown in Figures 2 and 4, respectively.

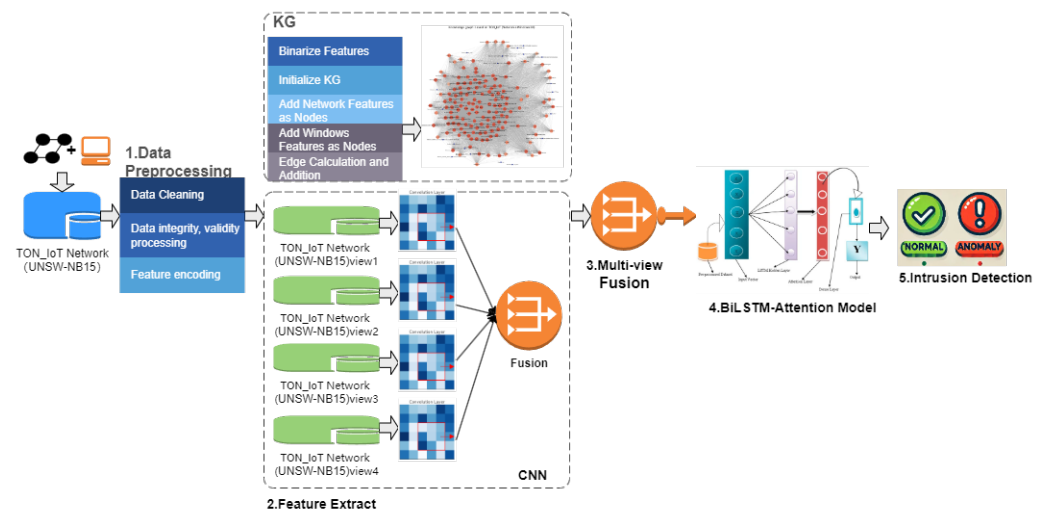


Figure 5. (KG+multi-CNNs)+attention LSTM multi-view model based on the TON_IOT network (UNSW-NB15).

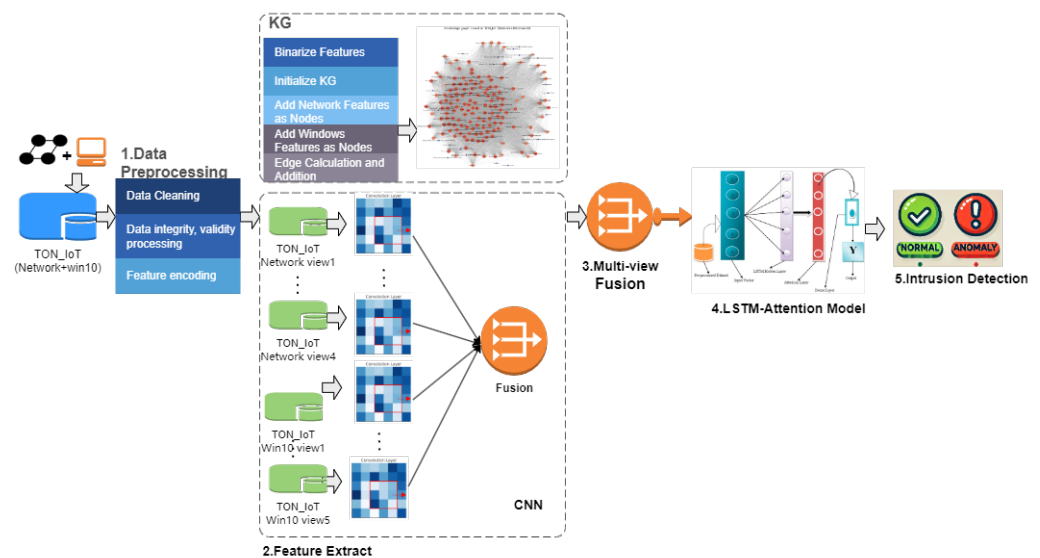


Figure 6. (KG+multi-CNNs)+attention LSTM multi-view model based on TON_IOT network+Win10.

3.3. Construction of the KG

This section provides a detailed explanation of an alternative feature fusion approach—one that is based on the KG. The KG aims to enrich the level of data representation by extracting relational features, thereby enhancing the model's capability to identify complex interactions. Therefore, as part of another fusion strategy, we construct a global KG to extract these relational features.

Depending on the characteristics of the different datasets, the construction process of the KG varies slightly. Figure 2 illustrates the construction process for the TON_IoT network and UNSW-NB15 datasets, while Figure 4 details the construction process for the combined TON_IoT network and Win10 dataset.

First, it is essential to understand some foundational definitions related to KG construction. The KG serves as structured semantic frameworks that effectively represent various concepts and the relationships connecting them within the real world [28]. Formally, a KG G can be described as $G = (E, R, T)$, where G is a labeled, directed multi-graph. In this context, $E = \{e_1, e_2, \dots, e_{|E|}\}$ represents the set of entities, while $R = \{r_1, r_2, \dots, r_{|R|}\}$ represents the set of relationships. The sizes of these sets are denoted by $|E|$ and $|R|$, respectively.

The facts represented within the KG are encoded as triples $T = \{(e, r, e') \mid e, e' \in E, r \in R\}$, which detail a specific relationship r linking the head entity e to the tail entity e' . Within a knowledge base, these facts are typically expressed in the form of triples like <entity, relationship, entity> or <concept, attribute, value> [25].

In this study, the construction of a KG mainly involves creating nodes and edges. Nodes represent the fundamental units within a KG and typically denote features or entities. In the scenario of network traffic analysis, nodes can be network features (such as traffic type and timestamp) or system features (such as Windows features). Edges represent the relationships between nodes. By effectively constructing nodes and edges, the KG can provide a comprehensive and enriched feature representation, thereby supporting subsequent feature fusion and anomaly detection. For the ontology design of the knowledge graph (KG), this study employs edge generation rules based on domain-driven feature classification and statistical constraints. Regarding semantic definitions, the approach relies on feature naming and dataset classification to directly represent domain semantics (see Section 4.1). The statistical analysis of edge relationships is not solely based on co-occurrence but also incorporates domain knowledge (see Algorithms 1 and 2). For the KG, we chose to use features as nodes and build edges based on co-occurrence because it lets us uncover meaningful relationships directly from the data without hardcoding them. Binarizing the edges keeps the graph simple and easier to work with. Grouping features by their source adds clarity to the structure and makes it easier to align with other parts of the model, like the CNN-based spatial features.

Feature extraction using the KG is divided into two steps: KG construction and knowledge extraction. The construction of the KG occurs during the data collection and preprocessing phase, where input data are loaded and feature identification is performed, including the extraction of both categorical and numerical features. Categorical features are encoded, and numerical features are normalized to ensure comparability across different feature types on a unified scale. Key features are identified through an evaluation of feature importance, followed by binarization to improve the model's ability to handle discrete data effectively. The co-occurrence relationships are obtained by binarizing the feature matrix and performing matrix multiplication ($X.T @ X$), where each element represents the number of times two features appear together within the same window. To ensure that the graph structure remains sparse and meaningful, a threshold is applied to the co-occurrence count or frequency, retaining only high-frequency feature pairs as edges in the KG. The importance of nodes in the graph is further quantified using eigenvector centrality, which supports subsequent model processing.

In the case of the TON_IoT network and Win10 combined dataset, which integrates two distinct data sources, it is necessary to add nodes and edges to these two sub-datasets separately when constructing the KG. Moreover, the KG must capture not only intra-source feature correlations but also inter-source interactions, adding complexity to the calculation of co-occurrence relationships between feature pairs. In the subsequent knowledge extraction and representation stage, node centrality is calculated to identify key nodes, which generally correspond to important features in network traffic analysis. The multi-level analysis of nodes and edges enables the resulting KG to effectively represent the intricate interactive relationships among features, supporting comprehensive feature fusion and anomaly detection.

The construction process of the KG is shown in Figure 2 (based on the TON_IoT network and UNSW-NB15 datasets) and Figure 4 (based on the combined TON_IoT network+Win10 dataset), covering the initialization of nodes, the generation of edges, and the calculation of the correlation of high co-occurrence features. Through these steps, the

KG effectively characterizes the relationships among network traffic features, laying the groundwork for the subsequent implementation of multi-view feature fusion.

Algorithm 1 provides a detailed description of the KG construction process for the combined TON_IoT network and Win10 dataset, while Algorithm 2 outlines the construction process for the TON_IoT network and UNSW-NB15 datasets.

Algorithm 1 KG Construction Based on TON_IoT network and Windows10 Dataset.

Require: Network dataset *binary_features_network_df*,

Windows dataset *binary_features_windows_df*

Ensure: KG *G*, Critical features tensors.

1. Initialize an empty graph *G* using NetworkX.

2. Add feature columns from each dataset as nodes in *G* with corresponding dataset attributes.

3. For each dataset, add edges based on feature co-occurrence:

Function ADD_EDGES_FROM_COOCCURRENCE(*df*, *G*, *dataset_name*, *threshold*):

Convert *df* to boolean matrix using threshold.

Compute co-occurrence matrix via boolean dot product.

Add edges for feature pairs with non-zero co-occurrence.

End Function

Call ADD_EDGES_FROM_COOCCURRENCE on both datasets.

4. Compute eigenvector centrality for all nodes.

5. Determine number of critical features to select from each dataset.

6. Sort nodes by centrality and select top features per dataset.

7. Convert selected feature names to tensor format:

Function STRING_TO_TENSOR(*feature_list*):

Convert strings to hashed tensor values.

End Function

Apply STRING_TO_TENSOR on selected features.

8. Visualize graph *G* using spring layout with node styling by centrality.

9. **Return** graph *G* and critical feature tensors.

Algorithm 2 KG Construction Based on TON_IOT Network Dataset (UNSW-NB15 Dataset).

Require: Training dataset *train_df*, Testing dataset *test_df*

Ensure: Preprocessed datasets, Feature co-occurrence KG *G*

1. Load training and testing datasets.

2. Detect numeric and categorical columns.

3. Encode categorical features using label encoding on combined unique values.

4. Normalize numeric features using MinMaxScaler fitted on training data.

5. Extract *X_train* and *y_train* from training dataset.

6. Compute mutual information between features and labels to select top *k* features.

7. Build co-occurrence matrix:

Binarize top features, compute co-occurrence counts for each feature pair.

8. Construct graph *G*:

Add feature nodes and edges for pairs with co-occurrence above threshold.

9. Visualize graph *G* with Kamada-Kawai layout, color/size based on edge weights.

10. **Return** preprocessed datasets and knowledge graph *G*.

4. Preparation for the Experiment

4.1. Data Set

In recent years, various datasets for intrusion detection have emerged for evaluating network security systems. However, many older datasets, such as NSL-KDD, do not accurately reflect the characteristics of modern, complex network attacks. Consequently, newer datasets have gained attention, including the CAIDA dataset [29], the ISCX dataset [30], CICIDS 2017 [31], the UNSW-NB15 dataset, and the TON_IoT, etc. [32,33]. In our study, we utilize three datasets—UNSW-NB15, TON_IoT network, and TON_IoT

network+Win10—for experimental deployment. The UNSW-NB15 dataset focuses on simulating traditional network environments, while the TON_IoT dataset emphasizes attacks on Internet of Things (IoT) devices and complex modern network environments. We also know that network data focus on external traffic characteristics, while host data focus more on internal activities and resource usage. Some malware may disguise itself as normal network traffic but behave abnormally in host behavior (such as attempting to modify critical system files). Therefore, we also deployed the experiment on a comprehensive data set that combines network activity data sets with host activity data sets, thereby trying to improve the model’s ability to detect complex attacks.

- (1) TON_IoT dataset: The dataset includes data on both normal operations and various intrusion events, aiming to closely simulate network behaviors observed in practical environments. In this study, we selected two types of data for combination: one is network traffic data, and the other is a dataset combining network traffic data and Windows data.
- (2) UNSW-NB15 dataset: The UNSW-NB15 dataset includes 2,540,044 records captured through an IXIA traffic generator. Following data cleaning and preprocessing, portions of the data were released as two distinct .csv files: the UNSW_NB15_training-set and the UNSW_NB15_test-set.

4.2. Multi-View Framework Integration

In this study, a multi-view framework was developed for each dataset to achieve effective multi-view feature fusion. The two recommended multi-view models are the simple model (CNN attention-based LSTM) and the complex model (multi-CNNs attention-based LSTM). The complex model involves dividing each dataset into multiple subsets via secondary feature fusion, extracting features from each subset independently, and subsequently combining them to achieve enhanced feature representation.

Tables 1–3 illustrate the details of the three datasets being divided into independent subsets. Notably, as TON_IoT network+Win10 is a combination of two datasets, when processing multiple views, the two datasets are divided into subsets for processing.

Table 1. Multi-view feature description of TON_IoT+Win10.

View Category	Features	Type
Processor Activity	Processor_DPC_Rate, Processor_pct_Idle_Time, etc.	Number
Network Activity	Network I/Current Bandwidth, Network I/Packets/sec, etc.	Number
Process Activity	Process_Pool_Paged Bytes, Process_IO Read Ops/sec, etc.	Number
Disk Activity	LogicalDisk(_Total)/Avg. Disk Bytes/Write, etc.	Number
Memory Activity	Memory/Pool Paged Bytes, Memory/Pool Non-paged Bytes, etc.	Number

The data categories are segmented based on the nature of the data and the specific types of information they capture during the inspection of network traffic. This segmentation is also applied to one-level fusion in the framework with two-level fusion. This idea supports the application of “multi-view feature fusion” in various modes.

Table 2. Multi-view feature description of UNSW-NB15.

View Category	Features	Type
Network Traffic	num_spkts, num_dpks, num_sbytes, etc.	Numerical
Temporal	num_dur, num_sinpkt, num_dinpkt, etc.	Numerical
Protocol and Service	cat_proto_3pc, cat_proto_a/n, etc.	Categorical
Security	num_sttl, num_dttl, num_sloss, etc.	Numerical
Connection	cat_state_CON, cat_state_ECO, etc.	Mixed (Mostly Numerical)

Table 3. Multi-view feature description of the TON_IoT network.

No.	Feature Category	Feature Name	No.	Feature Category	Feature Name
1	Intrinsic	src_ip	22	Content	ssl_cipher
2	Intrinsic	src_port	23	Content	ssl_resumed
3	Intrinsic	dst_ip	24	Content	ssl_established
4	Intrinsic	dst_port	25	Content	ssl_subject
5	Intrinsic	proto	26	Content	ssl_issuer
6	Intrinsic	src_bytes	27	Content	http_method
7	Intrinsic	dst_bytes	28	Content	http_uri
8	Intrinsic	src_pkts	29	Content	http_referrer
9	Intrinsic	dst_pkts	30	Content	http_version
10	Intrinsic	src_ip_bytes	31	Content	http_request_body_len
11	Intrinsic	dst_ip_bytes	32	Content	http_response_body_len
12	Content	service	33	Content	http_status_code
13	Content	dns_query	34	Content	http_user_agent
14	Content	dns_class	35	Content	http_orig_mime_types
15	Content	dns_type	36	Content	http_resp_mime_types
16	Content	dns_rcode	37	Time-based	network_ts
17	Content	dns_AA	38	Time-based	duration
18	Content	dns_RD	39	Host-based	conn_state
19	Content	dns_RA	40	Host-based	missed_bytes
20	Content	dns_rejected	41	Host-based	weird_name
21	Content	ssl_version	42	Host-based	weird_addl
			43	Host-based	weird_notice

5. Experiment and Results

5.1. Experiment Outline

In this section, the validation of the proposed multi-view fusion strategy for anomaly detection was undertaken through a series of comprehensive experiments, detailed as follows; then, all results are shown in Tables 4–8.

Table 4. Comparison of single-view and multi-view models based on the TON_IoT network+Win10 dataset.

Model Number	Model Type	Model Name	TON_IoT Network+Win10 (F1)
Model 1	Single-view	CNN	0.815
Model 2	Single-view	LSTM	0.726
Model 3	Single-view	CNN+Attn-Based LSTM	0.87
Model 5	Multi-view	(KG+CNN)+Attn-Based LSTM	0.95
Model 6	Multi-view	(KG+multi-CNNs)+Attn-Based LSTM	0.96

Table 5. The number of nodes and edges in the KG based on the three datasets.

Datasets	TON_IoT Network	TON_IoT Network+Win10	UNSW-NB15
Nodes	43	170	43
Edges	696	10,894	874

Table 6. Comparison of single-view and multi-view models based on the TON_IoT network dataset.

Model Type	Model Number	Model Name	TON_IoT Network (F1)
Single-view	Model 1	CNN [26,34]	0.7892/0.9111
	Model 2	LSTM [26,32]	0.8383/0.922
	Model 3	CNN+Attn-Based LSTM	0.93
Multi-view	Model 5	(KG+CNN)+Attn-Based LSTM	0.96
	Model 6	(KG+multi-CNNs)+Attn-Based LSTM	0.963

Table 7. Comparison of single-view and multi-view models based on the UNSW-NB15 dataset.

Model Number	Model Type	Model Name	UNSW-NB15 (F1)
Model 1	Single-view	CNN [27]	0.717
Model 2	Single-view	LSTM [27]	0.811
Model 3	Single-view	CNN+Attn-Based LSTM	0.814
Model 5	Multi-view	(KG+CNN)+Attn-Based LSTM	0.821
Model 6	Multi-view	(KG+multi-CNNs)+Attn-Based LSTM	0.837

Table 8. Ablation study results based on KG.

Model Name and Model Number	F1 Score		
	TON_IoT		UNSW-NB15
	Network	Network+Win10	
CNN+Attn LSTM (Model 3)	0.93	0.87	0.814
(KG+CNN)+Attn-Based LSTM (Model 5)	0.96	0.95	0.821
Multi-CNNs+Attn-Based LSTM (Model 4)	0.94	0.924	0.82
(KG+Multi-CNNs)+Attn-Based LSTM (Model 6)	0.963	0.96	0.837

5.1.1. Comparative Analysis of Baseline (Single-View) and Multi-View Models

In this analysis, we assessed the performance of the proposed multi-view models against the baseline single-view models. The results reveal that the multi-view models (KG+CNN and KG+multi-CNNs) demonstrate a significant improvement in anomaly detection accuracy by integrating relational and spatial features, thus outperforming the single-view models. This enhancement underscores the value of a multi-view approach in capturing complex feature interactions.

5.1.2. Comparison of Multi-View Fusion Strategies

To identify the most effective fusion strategy, we investigated two distinct multi-view fusion methodologies: KG+CNN and KG+multi-CNNs. Under consistent experimental conditions, these models comprehensively incorporate both host and network traffic characteristics, integrating spatial and relational information through one-level fusion (KG+CNN, Figures 1 and 3) and two-level fusion (KG+multi-CNNs, Figures 5 and 6), respectively. This comparative analysis aimed to discern which fusion strategy yields optimal performance across diverse datasets and anomaly detection tasks.

5.1.3. Ablation Study on KG Contribution

In previous experiments, we only used multiple views based on spatial features but ignored the relationship between features. This is also one of the main motivations of this study. A pivotal component of this research is the incorporation of relational features through the KG.

5.1.4. Generalization Evaluation Across Different Datasets

To further assess the generalizability of all proposed models across various network and host environments, ablation experiments on KGs are also included. We conducted the aforementioned experiments using three datasets: TON_IoT network+Win10, TON_IoT network, and UNSW-NB15. The F1 score served as the primary evaluation metric, with average values being computed over 10 repeated training iterations to ensure a robust and reliable performance evaluation.

All experiments were conducted on a computer equipped with an AMD Ryzen 7 5800U CPU (8 cores, 16 threads, 1.9–4.4 GHz) with integrated AMD Radeon Graphics, and additional hardware was provided by SRI, Center for High Performance Computing at Technological University of the Shannon, Midlands Midwest.

The software stack included: operating system: Windows 10 Pro (64-bit); Python: 3.8.10 (Anaconda distribution); deep learning framework: PyTorch 1.12.1; supporting libraries: scikit-learn 1.0.2, pandas 1.4.3, NetworkX 2.6.3.

5.2. Deployment of Single-View Models

This section describes the detailed deployment of experiments based on three single-view models (CNN, attention-based LSTM, and CNN+LSTM). The structure of the model has been introduced in the model design section of Section 3 (Sections 3.1–3.3).

5.2.1. Experimental Deployment Based on the TON_IoT Network+Win10 Dataset

Model 1: CNN. The dataset employed in this study comprises two components: network traffic data (Network) and host behavior data (Windows 10). To enable the effective integration of features from different sources, these two datasets were merged by concatenating them along the feature dimension, allowing the model to simultaneously leverage the complementary information inherent in each data source. Various preprocessing procedures were applied to standardize the datasets. For example: non-numeric features are encoded by LabelEncoder, and numeric features are standardized by StandardScaler. The architecture of the proposed CNN model consists of two convolutional layers containing 32 and 64 filters, respectively, each employing a convolutional kernel of size (3, 3). Batch-Normalization layers were utilized after each convolutional layer to accelerate convergence and stabilize training, followed by MaxPooling layers to reduce spatial dimensions and control overfitting. The final classification was carried out through a flattened layer followed by two fully connected layers. The model was compiled using the Adam optimizer ($\text{lr} = 0.0005$) and trained for 20 epochs, with a batch size of 32, optimizing for efficiency and generalizability.

Model 2: LSTM. The preprocessing approach adopted for the LSTM model mirrors that of Model 1, ensuring consistency across experiments. The LSTM model architecture comprises a single LSTM layer with an output dimensionality of 128, followed by a fully connected layer to perform the final classification. An attention mechanism was incorporated after the LSTM layer to enhance the model's ability to focus on key temporal features, thereby improving the overall representation of the feature set. During the training phase, the model was optimized using the Adam optimizer ($\text{lr} = 0.001$) and trained

for 20 epochs, using a batch size of 32, and a validation set ratio of 20% was employed to evaluate generalization capabilities during training.

Model 3: CNN+Attn-Based LSTM. In this hybrid model, spatial features are first extracted using a convolutional layer and subsequently passed to an LSTM layer for temporal feature extraction, with the LSTM layer being set to a hidden dimensionality of 128. An attention mechanism is added after the LSTM output. Finally, classification is performed through the fully connected layer, and the probability value is output using the Sigmoid activation function. The CNN model is a one-dimensional convolution layer that uses 32 filters and a convolution kernel size of three to extract spatial features. The kernel size of the maximum pooling layer (MaxPooling) is two, and the stride is two.

5.2.2. Experimental Deployment Based on the TON_IoT Network and UNSW-NB15 Dataset

Model 1: CNN. To verify the effectiveness of our proposed method, we compared the experimental results with the related methods in the reference [26,34]. Zhong Cao [26] implemented a single CNN model based on the TON_IoT network dataset. Asadullah Momand [34] compared ABCNN with traditional CNN when verifying the effectiveness of intrusion detection based on the TON_IoT network dataset. We also use them as the baseline model in this study to compare with our work. In a cited study [27], J. Vimal Rosyu used CNN as the baseline model to conduct intrusion detection experiments using the UNSW-NB15 dataset.

Model 2: LSTM. Zhong Cao [26] also introduced a single long short-term memory network (attention-based LSTM) model base on the TON_IoT network dataset enhanced by an attention mechanism. Raisa Abedin Disha [32] studied the impact of feature selection on model performance and used LSTM to process the ToN_IoT network dataset to solve the timing dependency problem in network intrusion detection. On the other hand, The attention-based LSTM model proposed in reference [27] was also employed to effectively capture the temporal dependencies present in network traffic data. Experiments conducted on the UNSW-NB15 dataset are revealed. Here, we use their experiments as a baseline model to compare with our subsequent experiments.

Model 3: CNN+Attn-Based LSTM. Building upon both datasets, a hybrid CNN and LSTM deep learning model was designed in this study for network traffic classification. The convolutional component of the model uses a kernel size of three for feature extraction, followed by a max pooling layer with a size of two for downsampling. The extracted features are then fed into a unidirectional LSTM module comprising 32 hidden units, which is responsible for capturing temporal dependencies. Finally, the binary classification task is completed through a fully connected layer. This integration aims to capitalize on the strengths of both CNNs and LSTMs—namely, spatial feature extraction and temporal pattern learning.

5.3. Deployment of Multi-View Models

Two multi-view models are recommended in this study, which perform one-layer fusion and two-layer fusion, respectively.

5.3.1. Experimental Deployment Based on the TON_IoT Network+Win10 Dataset

Model 5: (KG+CNN)+Attn-Based LSTM. This framework fuses the features extracted from KG and CNN, respectively, and further processes them through the long short-term memory network (LSTM) with an attention mechanism to achieve efficient feature extraction and classification (Figure 3). Given that the dataset is composed of network traffic and host data from the TON_IoT network and Windows 10 datasets, independent feature extraction processes were constructed for each data source. Specifically, during the construction of the comprehensive KG (Figure 4), nodes and edges were created for each

data subset independently. Nodes represent individual features, while edges denote feature pairs that co-occur based on specified threshold conditions. To digitize these co-occurrence relationships, binarization operations were applied, and edges were added both within the same dataset and across datasets according to predefined co-occurrence thresholds. The comprehensive KG thus constructed contains nodes derived from both the network dataset and the Windows dataset, with edges being formed through the calculation of co-occurrence relationships between feature pairs. Table 5 summarizes the number of nodes and edges created during KG construction process for this combined dataset.

For feature extraction, eigenvector centrality was utilized to identify the key features within each dataset. The main reason is that this method can quantify the importance of nodes in the global structure of KG and directly reflect the influence of features in the relationship network. Other methods, such as PCA, may destroy the interpretability of KG entity relationships and fail to preserve topological constraints. RFE has high computational cost, which is not suitable for our application scenario. Subsequently, features with the highest centrality scores were selected for the subsequent fusion process to ensure that the most influential features contribute to the model. The primary model parameters are configured as follows: A one-dimensional CNN (with an output channel size of 32 and a kernel size of 3) is applied to each dataset to extract 50-dimensional features. The LSTM layer uses a single-layer architecture with an attention mechanism, consisting of 128 hidden units. For training, the batch size was set to 16, and the learning rate was initialized at 0.0005. And the learning rate is optimized by the ReduceLROnPlateau scheduler.

The constructed comprehensive KG is shown in Figure 7. To distinguish nodes and edges corresponding to different data subsets, distinct color schemes were employed, as shown in Figure 8. Specifically, pink represents intra-dataset relationships, while orange represents inter-dataset relationships.

Model 6: (KG+multi-CNNs)+Attn-Based LSTM. This framework implements a secondary feature fusion process based on the initial fusion structure. Specifically, the whole process includes two levels of fusion (Figure 6).

In the first level, the TON_IoT network dataset and the Windows 10 dataset are each partitioned into multiple subsets (multi-views), with each subset undergoing independent feature extraction via distinct CNN units. Each CNN unit consists of a convolutional layer (with a kernel size of three and an output channel size of 32) followed by a fully connected layer. This step achieves the first fusion by integrating feature representations from different perspectives of the dataset.

In the second level of fusion, the multi-view features obtained from the first fusion are further combined with the critical relational features extracted from the KG to form a comprehensive feature representation. This dual-stage fusion aims to leverage both spatial characteristics captured through CNNs and relational information provided by KG, thereby enhancing the overall feature representation for anomaly detection.

The LSTM model within this framework is configured with 64 hidden units. Training is conducted using a binary cross-entropy loss function (BCEWithLogitsLoss), with the Adam optimizer being employed at a learning rate of 0.001. Table 5 provides the details of the nodes and edges generated during KG construction for the three datasets, offering insights into the underlying data relationships captured during feature fusion.

Knowledge graph based on TON_IoT (Network+Windows10)

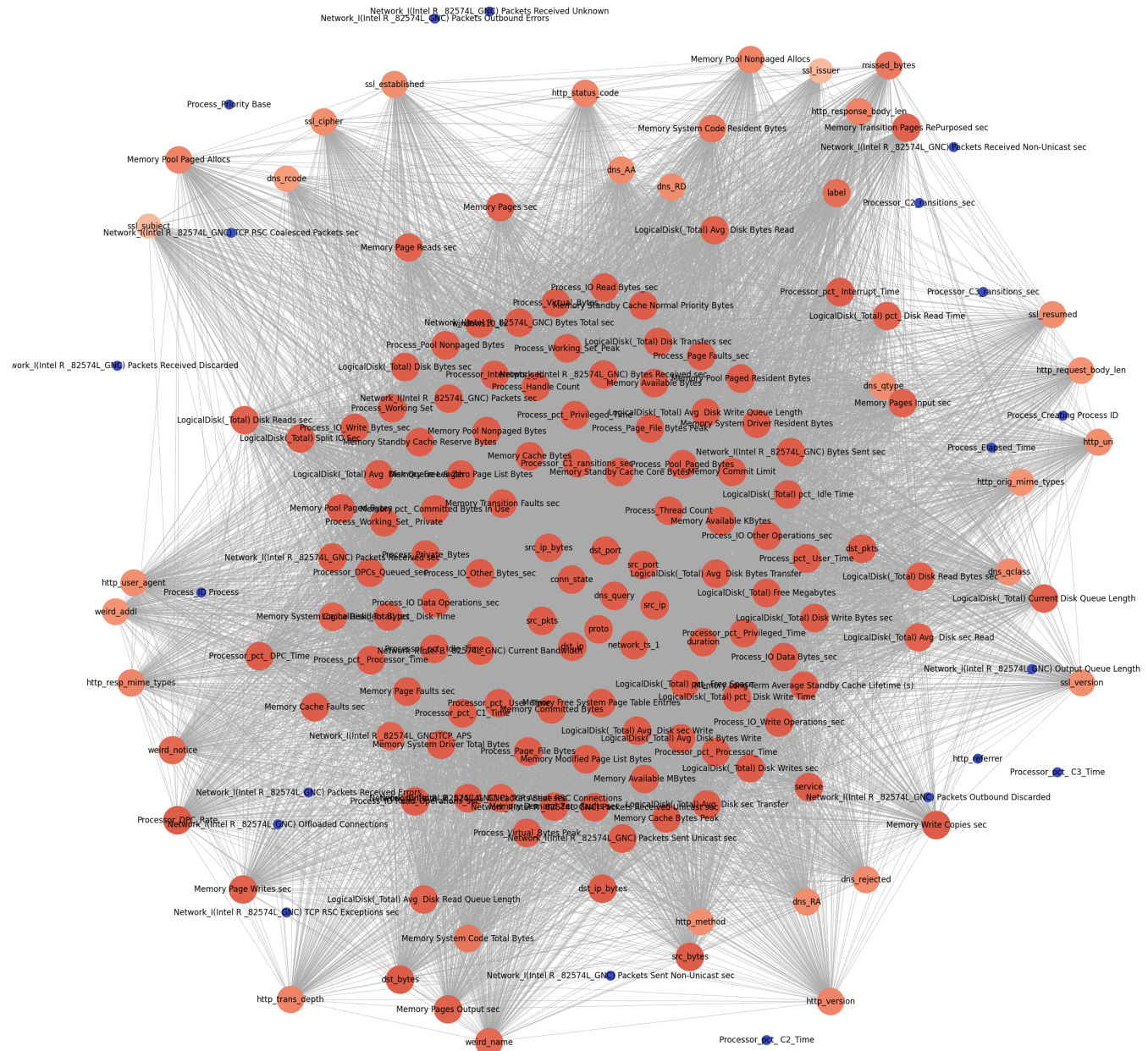


Figure 7. Knowledge graph based on the TON_IoT network and Windows 10.

The results are shown in Table 4, which is summarized in one table together with the single-view results obtained in Section 5.2.1 (TON_IoT network+Win10 dataset).

Knowledge graph based on TON_IoT (Network+Windows10)



Pink: relationships within the same dataset, orange: relationships between different datasets.

Figure 8. KG-built different datasets with different colors based on TON_IoT network+window10.

5.3.2. Experimental Deployment Based on TON_IoT Network and UNSW-NB15 Dataset

Within the multi-view framework, two distinct fusion models were deployed for the two datasets: Model 5 (Figure 3) and Model 6 (Figure 5). Both models extract relational features through the construction of KG (Figure 2), which are subsequently fused with spatial features obtained from convolutional layers.

Model 5 represents a one-level fusion framework within the multi-view architecture. Specifically, its CNN module utilizes a one-dimensional convolutional layer (with a convolution kernel size of 3×3) and a fully connected layer to extract spatial features, which are then fused with the relational features extracted from KG.

Model 6 implements a two-level fusion approach. Its feature extraction is completed by a multi-view CNN. Each view's CNN module consists of 64 filters with 3×3 convolutional

5.4. Deployment of Ablation Experiments

In the proposed multi-view fusion strategy, a KG is introduced to extract relational features, thereby enhancing the model's ability to capture complex feature relationships. To evaluate the impact of incorporating KGs on the performance of fusion strategies in both simple models (CNN-LSTM) and complex models (Multi-CNN-LSTM), we designed and conducted ablation experiments. Specifically, the ablation studies were carried out using the frameworks of the two recommended multi-view fusion strategies. The models compared are as follows:

Model 3 (without KG) vs. Model 5 (with KG): these comparisons were conducted to investigate the effect of KG integration on the simple CNN-attention LSTM architecture.

Model 4 (without KG) vs. Model 6 (with KG): these comparisons were aimed at assessing the contribution of the KG in more complex, multi-view fusion scenarios.

The architectural details of these four models are provided in Section 3, while the corresponding experimental deployment parameters are discussed in Section 5, Sections 2 and 3. In this section, we focus solely on presenting and analyzing the ablation experiment results to provide an intuitive understanding of the impact of the KG integration (see Table 8).

From the experimental results presented in Table 8, it is evident that the models incorporating the KG (Models 5 and 6) demonstrated improvements in F1 scores across all datasets. This observation further substantiates the important role of KG in capturing complex relationships among data and enhancing feature fusion. Notably, in the context of multi-view feature processing (Model 6), the integration of KG led to the optimization of the model's overall performance. These findings underscore the efficacy of KG in providing richer relational information, thereby facilitating a more comprehensive feature representation that ultimately enhances the model's ability to detect anomalies.

6. Conclusions

6.1. Experimental Conclusion Analysis

In alignment with the objectives of this study, this section presents a comprehensive comparison between multi-view fusion and single-view approaches in anomaly detection, as well as a comparison of fusion strategies. Particular emphasis is placed on verifying the effectiveness of incorporating KG as one of the views in the multi-view fusion framework. A thorough evaluation of the results from multiple perspectives is undertaken to ensure a robust understanding of the contributions of different methodologies. This section will provide a detailed analysis based on all the deployed experimental results.

6.1.1. F1 Score Analysis

1. Advantages of Multi-View Models

Across the three datasets, the F1 score of the multi-view model is higher than that of the single-view model. As presented in Table 4, within the TON_IoT network+Win10 dataset, the recommended multi-view approach (Model 6) achieved an F1 score of 0.962, representing a 7% improvement over the best-performing single-view model, which achieved an F1 score of 0.9.

The aforementioned experimental results were further corroborated using two additional datasets. As illustrated in Tables 6 and 7, the highest F1 scores obtained by the multi-view model (Model 6) for the TON_IoT network dataset and the UNSW-NB15 dataset were 0.973 and 0.837, respectively, indicating substantial improvements in both cases.

2. Comparison of Feature Fusion Strategies

The investigation into feature fusion strategies focused on two multi-view frameworks: (1) identifying the optimal fusion framework; and (2) evaluating the importance of KG-based relational features within the fusion strategy.

As depicted in Tables 4–7, the two-level fusion framework (Model 6) outperformed the primary fusion approach (Model 5). For instance, in Table 4, the F1 score of Model 6 for the TON_IoT network dataset was 0.973, representing an improvement over Model 5, which achieved an F1 score of 0.96. Similarly, the results presented in Tables 6 and 7 further reinforce the superiority of the two-level fusion strategy over the one-level fusion strategy.

The ablation experiment results presented in Table 8 further confirm the significance of incorporating KG-based relational features as an independent view in the fusion process, enhancing model performance. This enhancement is attributed to the effective use of complementary information from different sources, thereby enriching the representation of input features. For example, in the TON_IoT network+Win10 dataset, The F1 scores were 0.95 for Model 5 and 0.965 for Model 6, respectively—both outperforming the models that did not incorporate KG. These findings were similarly validated across the other two datasets (TON_IoT network dataset and UNSW-NB15 dataset).

6.1.2. ANOVA Summary Analysis

An analysis of variance (ANOVA) test was used to determine differences in mean F1 scores between multiple groups. In Figure 11, a violin plot combined with a honeycomb plot is used to illustrate the distribution of F1 scores for different models. In these plots, the width of the violin plot represents the density of data points, with the wider portion representing higher density. Each black dot represents the F1 score of a single experiment.

The results of the ANOVA test allow us to draw conclusions from three aspects:

1. *Superiority of Multi-View Models*

Across the three datasets, multi-view models—particularly Model 6—demonstrated superior performance in terms of higher F1 scores and reduced fluctuations. The F1 scores of the single-view models (Model 1 and Model 2) were notably lower than those of the multi-view models, underscoring that multi-view fusion enhances the generalization capability of the models across different datasets.

2. *Comparison of Feature Fusion Strategies*

The two-level fusion (Model 6) strategy outperformed the one-level fusion strategy (Model 5), as evidenced by both higher F1 scores and greater concentration of score distribution. This result highlights the effectiveness of the two-level fusion approach in synthesizing richer and more informative feature representations.

3. *Summary of Model Distribution Characteristics*

Model 6 not only exhibited superior mean F1 scores compared with other models but also demonstrated smaller variance, indicating more stable performance. The comparison in the ablation experiments (Model 3 vs. Model 5, Model 4 vs. Model 6) is illustrated in Figure 11. The models without KG features showed more dispersed distributions, thereby validating the performance gains achieved by incorporating KG features. Importantly, this performance improvement was consistent across all datasets.

Overall, the violin plots provide a visual summary of the significance and stability of the multi-view models, two-level fusion strategies, and KG-based feature fusion in enhancing model performance. The consistent results across different datasets further reinforce the robustness and effectiveness of these approaches.

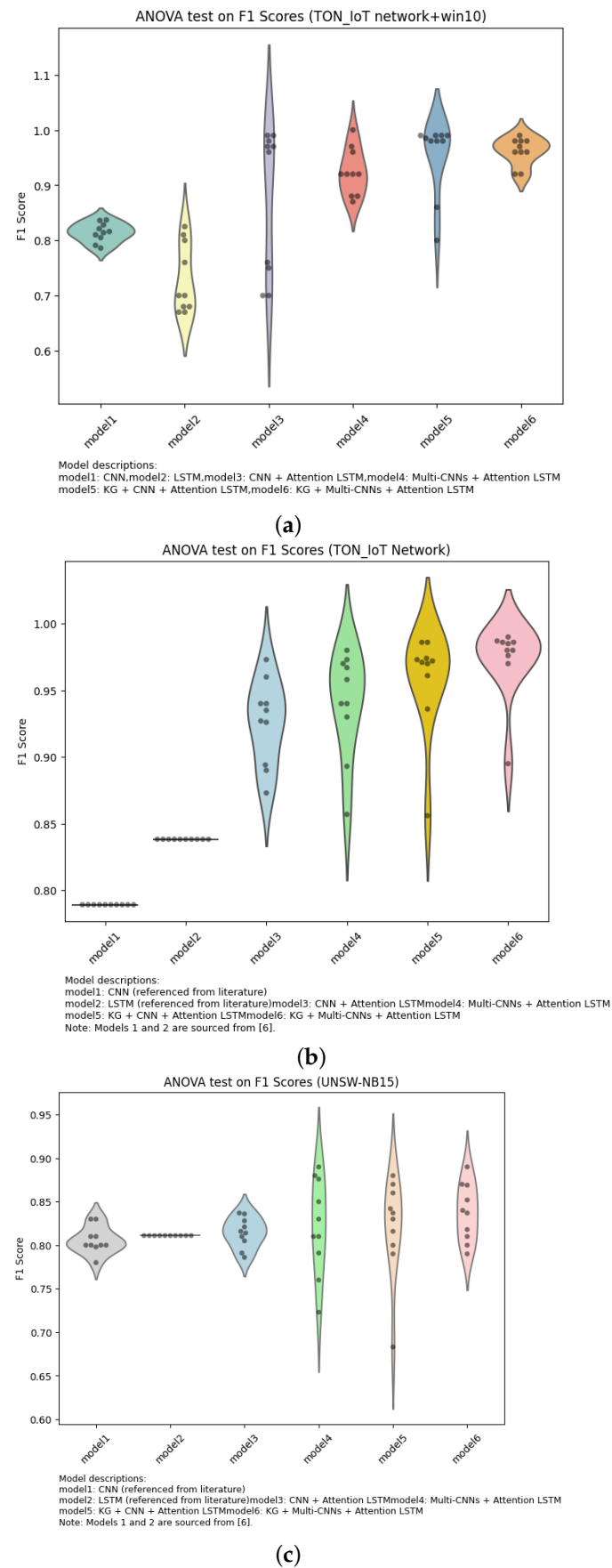


Figure 11. ANOVA test method based on three datasets: (a) TON_IoT network and Windows, (b) TON_IoT network, (c) UNSW-NB15.

6.2. Practical Application Value of the Model in Network Intrusion Detection

The practical application value of the proposed model can be demonstrated through several key aspects, as verified by the recommended approach in this study.

Firstly, the multi-view model effectively reduces false positives and false negatives when dealing with complex and diverse attack scenarios, thereby enhancing the reliability and robustness of intrusion detection systems. By integrating multiple perspectives of data, the model is better equipped to capture the nuances of both normal and anomalous behavior patterns, minimizing misclassification.

Secondly, the two-level fusion approach employed in the multi-source data fusion strategy enables a more comprehensive integration of the complementary information from each view. Its approach captures different layers of information, facilitating a richer feature representation.

The third aspect is the contribution of KG features. The KG helps the model better understand the complex relationships between network entities.

Although the proposed multi-view intrusion detection framework shows great potential, it still has some limitations. First, the framework assumes that the multi-view data input is relatively stable, which may not always hold true in dynamic or resource-constrained real-time environments. Second, the two-stage fusion imposes additional computational overhead. Third, building and maintaining an up-to-date knowledge graph in real time remains a challenging task.

In future work, we can improve in different areas based on the multi-view intrusion detection framework proposed in this study. **From a technical perspective of DL**, we can integrate cutting-edge deep learning progress from three directions, namely multimodal deep representation learning, knowledge reasoning enhanced by graph neural networks, and adaptive detection, to further improve system performance. **From the perspective of its application in real-time intrusion detection scenarios**, there is real-time multi-view data processing capability; real-time application value of the two-stage data fusion mechanism; and real-time auxiliary value of KG features. **In addition, there are also cross-domain fusion research directions**, such as migrating the multimodal adversarial training framework in medical imaging to network intrusion detection, building dynamic spatiotemporal graph neural networks, etc.

Author Contributions: M.L.: Conceptualization of this study, methodology, code, validation, writing—original draft, visualization. Y.Q.: conceptualization of this study, supervision. B.L.: funding acquisition, conceptualization of this study, methodology, writing—review and editing, supervision. All authors have read and agreed to the published version of the manuscript.

Funding: This publication has emanated from research conducted with the financial support of the Technological University of the Shannon under the President Doctoral Scholarship 2021; and the Horizon Europe Framework Program (HORIZON), under the grant agreement 101119681, Resilmesh.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: The dataset used in this study is free and publicly available on the Internet. We can obtain the TON IoT dataset and UNSW-NB15 dataset through the following link: <https://research.unsw.edu.au/projects/toniot-datasets> (accessed on 11 March 2025).

Conflicts of Interest: The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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